(R) Follow the Rules (1/5) [10 Points]

Languages change over time in systematic ways. For example, Old English had many words beginning with the letter **f** followed by **n**, which in Modern English has become **s**. Thus, Old English **fnesen** became Modern English **sneeze**. Luckily, the computational linguists at NACLO, Inc™ have discovered a machine in their basement which can model these changes. To do so, the machine takes **phonological rules** as input, which follow the pattern shown below:

f → s / #_n

This rule means: **f** becomes **s** in the following context: after the beginning of the word (**#**) and before the letter **n**. Here's another example of a rule, this time about the plural suffix **-s**. In earlier forms of English, this suffix was always pronounced **s**: this is still true for many words, like "lap**s**" or "leek**s**." However, in words ending with a **voiced**[1] sound (consonant or vowel), this suffix is actually pronounced **-z** (say, in "bags" or "lambs"). Writing this phenomenon down is a little more complicated, because many consonants can trigger it, so to model this using an algorithm, we must define the concept of **features**. Each sound can be positive or negative for a feature: for example, **p** and **t** are + for voicing (meaning that they have voicing), whereas **b** and **d** are - for voicing (meaning that they don't have voicing). The rule above then looks like this:

s → z / [+Voiced]_#

which means: **s** becomes **z** in the following context: after a voiced sound and before the end of the word.

Here's the list of English features we will use in Part 1:

+Const: all English consonants **+Vow**: all English vowels or sequences of vowels
+Voiced: voiced sounds (all vowels, plus b, d, g, j, m, n, l, v, w, y, z)
-Voiced: unvoiced sounds (ch, f, h, k, p, s, sh, t)

Finally, there is a special feature **Stop**. Sounds that are **+** for **Stop** come in pairs, with one member of the pair being **-Voiced** and the other **+Voiced**. Below, **p, t, k** are **-Voiced**, and **b, d, g** are **+Voiced** . The arrows indicate corresponding pairs; e.g., **p** is the unvoiced version of **b**.

+Stop: p ↔ b, t ↔ d, k ↔ g

[1] A voiced consonant is one that is pronounced with the vocal cords vibrating. You can test this by putting your hand on your throat as you say various consonants: when you say "m", "l", or "g", you will feel vibration in your throat, but "s", "t", "sh" won't produce this vibration. Make sure to do this quietly so that you don't bother the other NACLO contestants.

(R) Follow the Rules (2/5)

Running out of time to finish writing the NACLO contest, the NACLO team has discovered a time machine in their basement that they can use to get some more time. Unfortunately, they accidentally travel 300 years into the future, to the year 2324!

While stuck in the future, they encounter five new dialects of English, each of which uses different spelling conventions from modern-day English. Interestingly, it turns out that each dialect's spelling conventions can be related to modern-day spelling by one rule within the Phonological Machine. (Although the Phonological Machine was designed for handling rules about sounds, the linguists find that it is also effective for handling these spelling changes.)

Below are five sentences from the future dialects, as well as five rules that the linguists believe represent these dialects; however, the rules and the sentences are both in random order.

R1. On your Answer Sheets, indicate the correct correspondences between the rules and the sentences.

Dialect Rules

1. **[+Stop, +Voiced] → [+Stop, -Voiced] / _#**
   "All final stops become devoiced."
2. **[+Const]{2} → [+Const]{1} / _**
   "All doubled consonants become single consonants."
3. **s → z / #_**
   "All s's at the beginning of a word become z's."
4. **[+Vow] → Ø / [+Vow]_**
   "All vowels that follow another vowel are deleted."
5. **e → a / _[+Const]**
   "All e's before consonants become a's."

Sentences

a. "I am relly soked after my jorney to the floting osis."

b. "Aggplant fual has truly axcallant anargy afficiancy, far battar than an Earthling's alactric sockat."

c. "Can you realy believe things were so cheap back in 2251? Crazy that carots cost only 5 trilion dolars back then. It'd be a masive steal to get it for a quadrilion now."

d. "I neet to get a new robotic arm! Mine is so bat, ant I can't even use it in the tech-no-lap!"

e. "I zpilled my zynthetic zuperfood zyrup all over my zmart-zuit. Zigh."

<u>Make sure you record your answers in your Answer Sheets!</u>

(R) Follow the Rules (3/5)

The NACLO team have finally gotten their time machine working again, but now they've accidentally traveled some 1,600 years into the past, to the Roman Empire! All is not lost, though: the NACLO team has a chance to learn about **Vulgar Latin**, the Latin spoken by common people in the late Roman era, and that is the ancestor of today's Romance languages, like Portuguese, Italian and Sardinian. On the next page are some data they manage to collect.[2]

After returning back to the year 2024, the NACLO team plug their data back into the Phonological Machine. Some computation later, it spits out a most likely family tree, shown at the end of this problem (page 5/5).

R2. On your Answer Sheets, identify Languages A, B, and C.[3]

R3. Fill in blanks (1) - (23) for the NACLO Phonological Machine. Note that each blank can have one or more characters. See the **Usable Features** and **Notes** below for further helpful information.

R4. Provide the Western Romance descendant of the following Latin words: BACULUM, VĪTA, FINDĀRE.

Usable Features:

[+Const]: b,c,d,f,g,h,l,m,n,p,r,s,t,v

[+Vow]: a,e,i,o,u

[+Stop]: b,c,d,g,p,t

[+Stop +Voiced, +Stop -Voiced] pairs: b ↔ p, t ↔ d, c ↔ g

[+Nasal, -Nasal] pairs: a ↔ ã, e ↔ ẽ, i ↔ ĩ, o ↔ õ, u ↔ ũ

[+L, -L] pairs: a ↔ ā, e ↔ ē, i ↔ ī, o ↔ ō, u ↔ ū

Notes:

- The Portuguese writing system roughly represents Old Galician-Portuguese as of the year 1200. The Italian writing system roughly represents Italian as of the year 1500. The Sardinian writing system roughly represents Sardinian as it is spoken today, with one correction to account for two letters that are pronounced the same.
    - If you know Portuguese, you might note some of the words look a bit different from what you're used to seeing: that's because the words as written are from Old Galician-Portuguese, the ancestor of Portuguese.
- In Latin, the macron (¯) over a vowel represents a long vowel. **C** is always hard: **centum** ("100") is pronounced with a hard "k" at the beginning of the word.
- In Portuguese, the tilde (˜) over a vowel represents that the vowel is nasalized (pronounced with air going through the nose).

[2] In real life, we have very little record of what Vulgar Latin was like: our main sources come from popular writing such as graffiti, complaints by grammarians about common grammar mistakes, and loans into non-Romance languages. The above data represents linguists' best guess at Vulgar Latin.

[3] Knowledge of Latin, Portuguese, Italian, or Sardinian is not necessary or helpful to solve the problem.

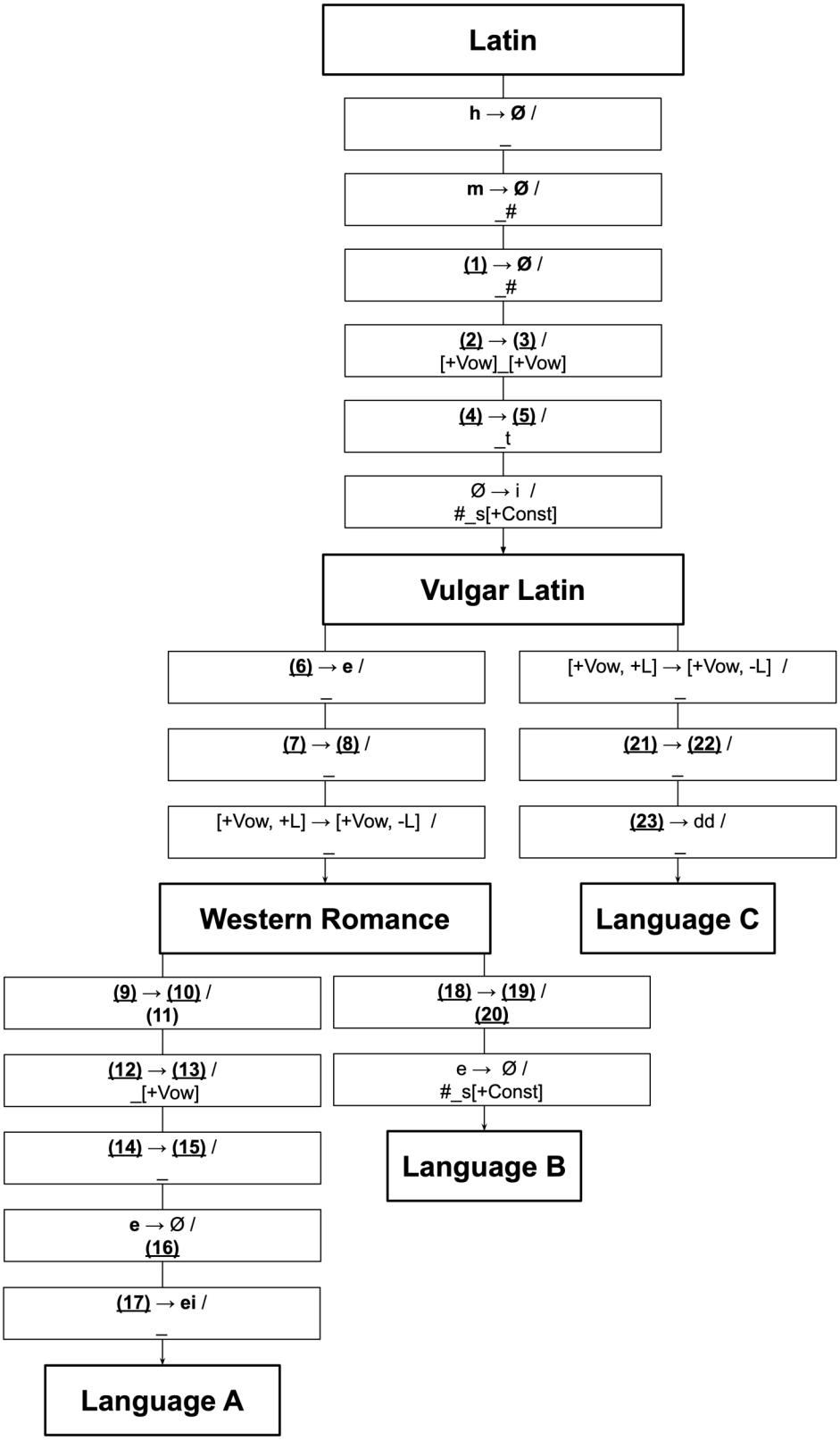
(R) Follow the Rules (4/5)

English	Latin	Vulgar Latin	Portuguese	Italian	Sardinian
"fireplace"	FOCUM	*focu*	fogo	fuoco	focu
"good"	BONUM	*bonu*	bõo	buono	bonu
"broken"	RUPTUM	*ruttu*	roto	rotto	ruttu
"horse"	CABALLUM	*cavallu*	cavalo	cavallo	cabaddu
"to play (inst.)"	SONĀRE	*sonāre*	sõar	suonare	sonare
"dust"	PULVERA	*pulvera*	*(no cognate)*	polvera†	pulbera*
"snow"	NIVIS	*nivi*	neve	neve	nibi*
"pen"	PENNA	*penna*	pena	penna	*(no cognate)*
"fold"	SINUM	*sinu*	seio	seno	sinu
"to rub"	FRICĀRE	*fricāre*	fregar	frecare†	fricare
"choose"	ADOPTĀRE	*adottāre*	adotar	adottare	*(no cognate)*
"green"	VIRDEM*	*virde*	verde	verde	birde*
“arena”	ARĒNA	*arēna*	areia	arena	arena
"fodder"	CIBUS	*civu*	cevo	*(no cognate)*	cibu
"to have"	HABERE	*avere*	aver*	avere	abere
"vain"	VANUM	*vanu*	vão	vano	*(no cognate)*
"dry"	SICCUM	*siccu*	seco	secco	siccu
"skin"	PILUM	*pilu*	pelo	pelo	pilu
"to send"	MITTERE	*mittere*	meter	mettere	mittere*
"estate"	VĪLLA	*vīlla*	vila	villa	bidda
"loin"	LUMBUM	*lumbu*	lombo	lombo	*(no cognate)*
"state"	STATUM	*istatu*	estado	stato	istatu
"spatula"	SPATHA	*ispata*	espada	spata*	ispata*
"pear"	PIRA	*pira*	pera	pera	pira
"young chick"	PULLUM	*pullu*	polo	pollo	puddu
"castle"	CASTELLUM	*castellu*	castelo	castello	casteddu
"husband"	MARĪTUM	*marītu*	marido	marito	*(no cognate)*
“moon”	LŪNA	*lūna*	lũa	luna	luna

† represents older spellings that have since changed. * represents words that we have modified the spelling of for regularity. You can ignore these symbols. Also, some Latin words have no descendants in one of the daughter languages, and are marked (no cognate).

<u>Make sure you record your answers in your Answer Sheets!</u>

(R) Follow the Rules (5/5)



Make sure you record your answers in your Answer Sheets!

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Answer Sheets

REGISTRATION NUMBER					

Name: _____

Contest Site: _____

Site ID: _____

City, State/Province: _____

Grade: _____

Please also make sure to **write your registration number and your name on each page of the Answer Sheets**, and **turn in all pages of the Answers Sheets** even if you have left some blank .

SIGN YOUR NAME BELOW TO CONFIRM THAT YOU WILL NOT DISCUSS THESE PROBLEMS WITH ANYONE UNTIL THEY HAVE BEEN OFFICIALLY POSTED ON THE NACLO WEBSITE IN APRIL.

Signature: _____